DEVELOPMENT OF A LOCALIZED REAL-TIME FREQUENCY MONITORING SYSTEM FOR STRUCTURAL HEALTH MONITORING USING MEMS IMUS AND ESP32

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APPROVAL SHEET

This thesis titled **"Development of a Localized Real-Time Frequency Monitoring System for Structural Health Monitoring using MEMS IMUs and ESP32"**, prepared and submitted by **JAN CLYDE T. TALOSIG** in partial fulfilment of the requirements for the degree, Bachelor of Science in Electronics Engineering, is hereby recommended for final defense.

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DEDICATION

This section is a personal note where the researcher dedicates the work to individuals or groups who provided inspiration, support, or motivation. It is usually written in a heartfelt and informal tone compared to the rest of the manuscript.

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This section expresses gratitude to those who contributed to the completion of the study. It can include advisers, faculty members, classmates, family, friends, and institutions. The tone is formal but appreciative.

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ABSTRACT

TITLE : DEVELOPMENT OF A LOCALIZED REAL-TIME FREQUENCY MONITORING SYSTEM FOR STRUCTURAL HEALTH MONITORING USING MEMS IMUS AND ESP32

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This developmental study aims to develop and field-evaluate a localized system for recording roadway-deck vibration-frequency indicators on Buntun Bridge in Tuguegarao City, Cagayan. The preliminary design uses one PCB carrying an ESP32-S3 and three co-located triaxial MPU-6050 IMU ICs. The board will be secured at one predeclared roadway-deck station, subject to survey, safe mounting, and site authorization. Each field session will last 10 minutes while a separate observer tallies vehicles at a safe viewpoint. Pilot counts will define and freeze the local NEAR_ZERO, MEDIUM, and HEAVY groups before formal collection; these labels are not universal traffic classes. Vibration records will be divided into pilot-checked 60-second analysis windows. Arduino IDE firmware will poll the three channels sequentially, timestamp X/Y/Z readings, and apply mean removal, a fixed window, FFT magnitude, and the largest positive-frequency bin within a predeclared band. Valid window estimates will be summarized per session, channel, and axis. The mean and sample standard deviation will describe baseline spread; a separate two-standard-deviation band will prompt review only. The study evaluates acquisition, repeatability, and visualization; it does not diagnose damage or assess bridge safety. No field results are yet available.

Keywords: Buntun Bridge; roadway-deck vibration; dominant spectral frequency; FFT; MEMS IMU

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Bridges are critical components of transportation infrastructure, serving as lifelines for economic activity, emergency response, and regional connectivity. Ensuring their safety and long-term performance is essential, particularly in the Philippines, which is highly exposed to natural hazards such as earthquakes, typhoons, flooding, and soil erosion. These environmental and operational stresses accelerate structural deterioration, making continuous monitoring increasingly important (Brownjohn, 2007; Farrar & Worden, 2007).

On February 27, 2025, part of the newly completed Cabagan–Sta. Maria Bridge in Isabela collapsed while a heavily loaded dump truck was crossing it. Contemporary reporting recorded six injuries, and the Department of Public Works and Highways said it was assessing the cause (Department of Social Welfare and Development, 2025; Philippine News Agency, 2025a).

On October 6, 2025, the Piggatan Bridge in Alcala, Cagayan, also collapsed while heavy trucks were crossing. The Department of Public Works and Highways dispatched a team to assess the damage. These reports provide regional context; they do not show that this study's system could have predicted the collapse (Philippine News Agency, 2025b).

Buntun Bridge, opened in 1969 and spanning 1,369 meters, remains a direct route between Tuguegarao City and Solana. In March 2025, DPWH restricted trucks weighing more than 15 tons from crossing. This study uses the bridge as a field site for recording

roadway-deck vibration-frequency behavior under naturally occurring traffic; it does not assess the bridge's safety or load capacity (Philippine Information Agency, 2025; Philippine News Agency, 2025c).

These events show the consequences of bridge failures and the need for careful infrastructure management. They do not establish that frequency monitoring could have predicted or prevented either collapse. The present study therefore addresses a narrower task: recording and characterizing roadway-deck vibration-frequency behavior at a fixed Buntun Bridge station under observed traffic conditions.

Structural Health Monitoring (SHM) has emerged as a modern engineering approach for assessing the condition of structures through continuous or periodic measurement of structural responses (Brownjohn, 2007; Farrar & Worden, 2007). Among various SHM techniques, frequency-based monitoring is widely recognized as a method for observing structural dynamic changes (Avci et al., 2021; Azhar et al., 2024). The natural frequencies of a structure depend on its mass and stiffness; therefore, damage such as cracking, corrosion, or joint degradation can alter frequency characteristics (Salawu, 1997; Doebling et al., 1996; Carden & Fanning, 2004). Vibration signals can be transformed for frequency analysis: the Fast Fourier Transform (FFT) computes the discrete Fourier transform efficiently, while Welch's method estimates power spectra by averaging modified periodograms (Cooley & Tukey, 1965; Welch, 1967). A selected peak is still an observed spectral feature whose interpretation depends on the signal and measurement conditions (Hasani & Freddi, 2023; Rainieri & Fabbrocino, 2014).

Published bridge-monitoring studies describe varied sensor, data-acquisition, embedded-processing, and communication architectures, including low-cost MEMS-based field systems (Komarizadehasl et al., 2022a; Hidalgo-Fort et al., 2024; Giermann et al., 2025). Measurement limitations and deployment conditions need to be evaluated for each implementation. This study develops and field-evaluates a localized prototype using MEMS IMUs, an ESP32-S3, Arduino IDE firmware, and web visualization on Buntun Bridge. It characterizes dominant spectral-frequency indicators under the observed NEAR_ZERO, MEDIUM, and HEAVY traffic categories; it does not evaluate existing inspection practices or claim to replace structural inspection.

To address this applied development need, this study proposes a localized real-time frequency-monitoring system using one PCB with an ESP32-S3 and three co-located triaxial MEMS IMU ICs. The single board will be mounted at one fixed roadway-deck station. Arduino IDE firmware will poll the three channels sequentially, process each channel-axis record using the thesis's FFT peak-selection method, and send session summaries to a web interface. A concurrent visual vehicle tally will describe the traffic observed during fixed-duration sessions. The system will characterize local dominant spectral-frequency indicators without claiming a bridge-wide modal survey, structural diagnosis, or explanation of previous bridge failures (Cooley & Tukey, 1965; Gaviña et al., 2017; Payawal et al., 2017; Lau et al., 2018).

1.2 Statement of the Problem

Recent bridge failures in Cagayan Valley underscore the consequences of infrastructure disruption, but they do not define the technical problem measured in this

study. The research problem is how to develop a low-cost system that can acquire and process acceleration records from a predefined roadway-deck station on Buntun Bridge and establish an internal baseline of dominant spectral-frequency indicators under observed traffic-density conditions. The system is intended to supplement frequency observation; it is not designed to detect structural stress, determine load limits, or predict failure. Specifically, the study seeks to answer the following questions:

1. How can a single-board system acquire and process complete, timestamped X/Y/Z acceleration records from three co-located IMU channels at the predefined roadway-deck station?
2. How consistently can the system produce per-channel, per-axis session estimates across the three locally defined traffic-count groups?
3. How can the web-based interface display per-sensor, per-axis frequency estimates, traffic category, baseline, and review status?

1.3 Objective of the Study

This study aims to develop and field-evaluate a localized real-time frequency-monitoring system for roadway-deck vibration on Buntun Bridge using one ESP32-S3 PCB with three co-located triaxial MEMS IMUs. Specifically, it aims to:

1. Develop a single-board system that sequentially acquires timestamped X-, Y-, and Z-axis readings from each onboard IMU channel at the fixed roadway-deck station, and evaluate record completeness and timing;

2. Extract dominant spectral-frequency indicators from fixed 60-second windows, summarize them by 10-minute session, and construct descriptive baselines for each traffic group, onboard channel, and axis; and,
3. Develop a web-based interface that displays logged per-channel, per-axis results, session traffic count and group, baseline, and review-only status.

choose one only

1.4 Conceptual or Theoretical Framework of the Study

In structural dynamics, modal frequencies depend on structural mass and stiffness; changes in physical properties may therefore change modal behavior. The dominant peak in an operational roadway-deck vibration record, however, is a local observed spectral indicator. Excitation, roadway-surface condition, environmental and operational conditions, sensor characteristics, and measurement noise can also change measured peaks. Accordingly, this study treats a peak shift as a change in the recorded indicator that warrants review, not as proof of a particular damage mechanism or structural condition (Sohn, 2007; Hasani & Freddi, 2023; Camara et al., 2017).

The conceptual framework of the developed system follows an Input–Process–Output (IPO) model, illustrated in Figure 1.

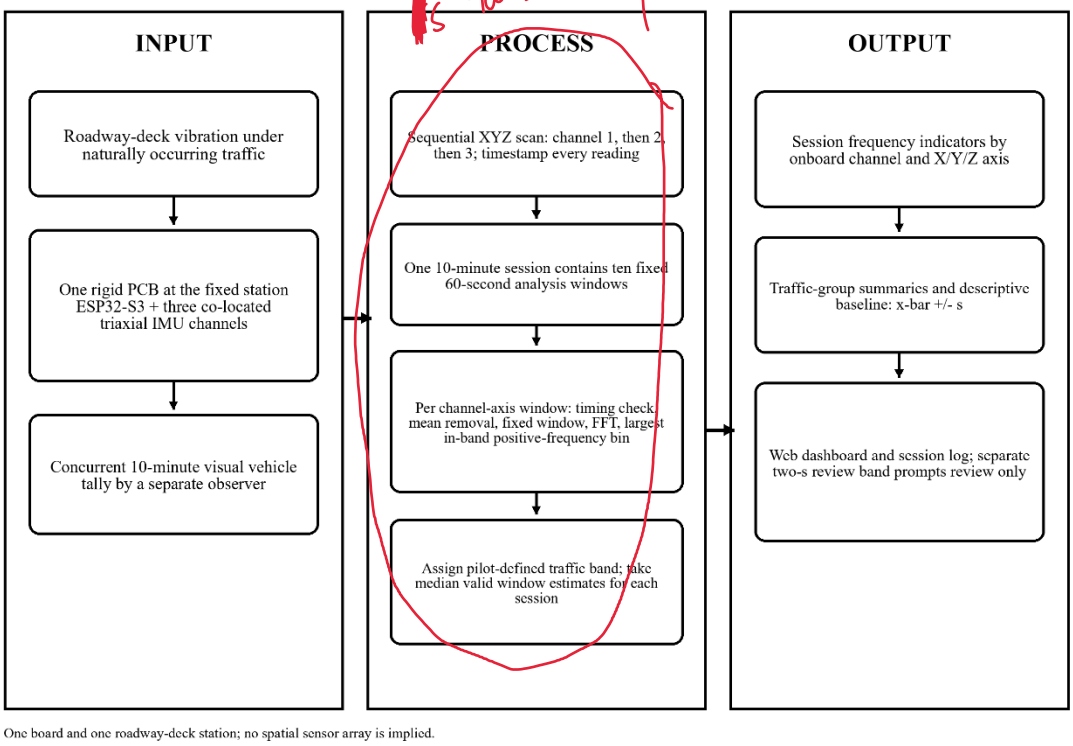


Figure 1. Input–Process–Output (IPO) Conceptual Framework of the SHM System

In the Input stage, naturally occurring traffic and other ambient excitation produce roadway-deck vibration at the predefined Buntun Bridge station. One rigid sensing PCB carrying three co-located triaxial IMUs records that local response. A separate observer tallies vehicle passages during the same 10-minute session.

During processing, the ESP32-S3 polls onboard IMU channels in the fixed sequence 1, 2, 3; each reading is tagged by channel, axis, sample index, timestamp, and scan-cycle ID. The session is divided into ten 60-second analysis windows. Each channel-axis window is timing-checked and processed using the fixed FFT settings. The system associates its valid window estimates with the session's pilot-defined traffic-count group and compares

the session summary with the matching baseline; a value outside the review band is flagged for human review only.

In the Output stage, per-session roadway-deck frequency indicators, vehicle-count context, and review status are transmitted to a web-based interface and retained in the session log.

This framework concerns one local roadway-deck station and the development of its acquisition, frequency-extraction, and visualization pipeline. The three onboard channels are co-located on one rigid board; the study does not estimate spatial mode shapes or claim to measure the bridge's global response.

1.5 Significance of the Study

This study contributes a developmental prototype for localized roadway-deck vibration-frequency monitoring in the Philippines. One PCB with three co-located triaxial MEMS IMUs and an ESP32-S3 will acquire and process vibration data for web-based display. Its field reliability, long-term performance, and suitability as an alternative to existing SHM systems remain to be evaluated.

To residents and commuters in Tuguegarao City and Solana, this study may provide a documented record of Buntun Bridge roadway-deck vibration-frequency behavior at the selected station under the observed NEAR_ZERO, MEDIUM, and HEAVY traffic categories. The prototype is a research instrument; it does not determine whether the bridge is safe or replace inspections, load restrictions, maintenance decisions, or emergency management.

For engineers and infrastructure inspectors, the prototype is intended to provide logged roadway-deck vibration-frequency observations that may supplement visual inspections. This study does not validate an automated condition assessment, maintenance decision rule, or change to inspection intervals.

For DPWH and other government agencies responsible for bridge maintenance, this study documents a prototype data-acquisition and display workflow designed to establish an empirical, site-specific roadway-deck frequency baseline. The measurements may support future monitoring research, but this study does not validate a maintenance decision rule, assess structural safety, or justify changes to inspection intervals.

For the local economy and communities in Region 2, this thesis documents development of a prototype intended to record Buntun Bridge roadway-deck vibration-frequency behavior under observed traffic. The study does not evaluate whether using this system could prevent bridge closures or reduce economic disruption.

For the academic and engineering research community, this thesis documents a localized single-board prototype and field procedure for collecting and summarizing roadway-deck vibration-frequency indicators at one fixed Buntun Bridge station using three co-located triaxial IMU channels, embedded FFT processing, and web visualization. Future work may evaluate other structures, spatial sensor arrangements, or damage-related indicators; these are outside this study.

The prototype uses locally accessible components as a developmental approach to bridge-frequency monitoring. Its cost, reliability, and scalability will need to be evaluated before broader deployment can be claimed.

1.6 Scope and Delimitation

This study develops and evaluates a localized frequency-monitoring system using one PCB carrying an ESP32-S3 and three co-located triaxial MEMS IMU ICs. The single board will be installed at the predefined roadway-deck station shown by the red marker in Figure 2. The external field supply connects through the board's USB-C receptacle, which also provides the ESP32-S3 USB programming connection; field summaries use the separate wireless path. Firmware polls channel 1, channel 2, and channel 3 in that order. Timestamp and scan-cycle fields support traceability; the readings are sequential, not simultaneous, and the co-located channels do not form a spatial bridge array.

The system will be tested on Buntun Bridge in Tuguegarao City, Cagayan, at the mapped roadway-deck station shown in Figure 2. The red marker is an approximate map reference, not a surveyed coordinate. The study's performance and extracted frequency characteristics are specific to this single site and the documented station; they are not claimed to generalize to bridges of different structural types, spans, or material conditions without further testing.

Since no prior instrumented baseline exists for Buntun Bridge, this study establishes its baseline frequency ranges using the developed device itself, based on vibration data gathered under varying vehicular traffic conditions. The resulting baseline is therefore internal to the system and this deployment, and is not validated against an external reference sensor or a separately verified ground truth.

The study is limited to frequency-based monitoring and does not include structural diagnosis, damage classification, predictive modeling, load-capacity analysis, or repair

recommendations. Interpretation of any observed frequency shift as a specific damage type or severity level is outside the scope of this study.

The system is further limited by hardware constraints such as processing speed, memory capacity, sensor resolution, sequential acquisition, and the fact that the three IMUs are co-located on one PCB. Environmental factors such as temperature, humidity, wind, electromagnetic interference, roadway-surface condition, and vehicle-pavement interaction are not fully incorporated into the analysis; relevant conditions will be recorded as context when practicable. Data collection is also limited to the specific station, dates, and traffic sessions observed and will not capture the full range of seasonal, weather-related, spatial, or long-term structural variation.

make it BOLD

1.7 Definition of Terms

Baseline Frequency Range: The descriptive $\bar{x} \pm s$ spread of dominant spectral-frequency estimates for a specified traffic category, sensor, and axis, used as an internal reference for later observations from the same stratum.

Excitation: An external force or disturbance, such as vehicular traffic, wind, or ambient vibration, that causes a structure to vibrate and produce a measurable dynamic response.

ESP32-S3*:* A microcontroller used for data acquisition, processing, and wireless communication in the system.

Fast Fourier Transform (FFT)*:* A signal processing technique used to convert time-domain signals into frequency-domain data.

Frequency Monitoring: The process of analyzing roadway-deck vibration signals to extract dominant spectral-frequency indicators. An FFT-selected peak in this study is not automatically identified as a natural frequency or structural mode.

Dominant Spectral-Frequency Indicator: The frequency of the largest magnitude positive-frequency FFT bin within the predeclared analysis band for one sensor-axis record; it is a measured spectral feature and is not automatically a verified natural frequency or mode.

Localized (System)*:* In this study, refers to a system designed using cost-efficient, locally accessible components rather than imported, proprietary hardware, making it more feasible to deploy and maintain in resource-limited settings.

MEMS IMU (Micro-Electro-Mechanical Systems Inertial Measurement Unit): A device containing inertial sensors. In this study, each MPU-6050 IC provides triaxial acceleration and angular-rate channels; the frequency analysis uses acceleration data from three co-located ICs on one PCB.

Natural Frequency: The inherent frequency at which a structure vibrates in a particular mode when disturbed, determined by its physical properties and boundary conditions. An operational roadway-deck spectral peak is not automatically equivalent to a verified natural frequency.

Real-Time Monitoring*:* Continuous or near-instantaneous observation and processing of data.

Resonant Frequency*:* The frequency at which a structure experiences maximum vibration amplitude due to external excitation.

Structural Health Monitoring (SHM): A method of evaluating the condition of a structure through measurement and analysis of its response.

Vibration Signal: The raw time-domain data captured by the sensor, representing the structure's motion or acceleration over time, prior to frequency-domain processing.



Traffic-count group: A study-local label assigned from the total number of vehicles counted during one fixed 10-minute session. NEAR_ZERO denotes the lowest pilot-observed count band; MEDIUM and HEAVY denote the next two bands. The names are not universal thresholds and must be tied to the frozen local cutoffs.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter reviews literature and related studies relevant to the development of a localized, real-time frequency monitoring system for bridge Structural Health Monitoring (SHM). The review is organized thematically to connect the study with established work on vibration-based SHM, structural dynamic properties, operational modal analysis, frequency-domain signal processing, low-cost MEMS sensing, real-time and Internet of Things (IoT)-enabled monitoring, and the effects of environmental and operational variability. International and Philippine studies are considered to identify established practices, practical limitations, and the specific gap addressed by the present study.

2.1. Introduction

Structural Health Monitoring has developed from periodic condition assessment toward measurement-based approaches capable of observing structural response during service. For bridge structures, vibration measurements are particularly useful because dynamic behavior can be observed without requiring the application of a controlled test load. Contemporary reviews describe bridge SHM as a combination of sensing, data acquisition, signal processing, modal identification, and decision-support methods that can

operate continuously or periodically depending on the monitoring objective (Azhar et al., 2024; Chaudhari et al., 2026). The literature therefore provides a technical basis for a field-deployed system that measures bridge vibration, converts the response into frequency-domain information, and presents the extracted parameters for monitoring.

The reviewed literature also shows that frequency-based monitoring is not a simple matter of obtaining a spectral peak and treating every change as damage. Modal frequencies are influenced not only by structural condition but also by temperature, humidity, wind, traffic loading, boundary conditions, sensor characteristics, and measurement noise (Sohn, 2007; Wang et al., 2022). This distinction is important for the present study because the intended baseline is to be established on an operating bridge under different vehicular traffic conditions. The literature consequently supports the use of repeated observations and explicit characterization of operational variability rather than a single reading taken under one field condition.

2.2. Vibration-Based Structural Health Monitoring of Bridges

2.2.1 Structural Health Monitoring and Dynamic Response

Structural Health Monitoring refers to the systematic observation of a structure through measured responses and the analysis of those responses to identify changes in behavior, condition, or performance. Vibration-based SHM is one of the principal approaches because structural vibration carries information about dynamic properties that depend on the physical characteristics of a structure. Reviews of vibration-based damage detection show that changes in modal properties, including natural frequencies and mode

shapes, have long been used as damage-sensitive features because alterations in mass, stiffness, or boundary conditions can modify the dynamic response (Das et al., 2016; Avci et al., 2021).

For bridges, vibration monitoring is particularly suitable for in-service observation because ambient and operational excitation from traffic, wind, and other sources can provide measurable structural response without the need for a dedicated excitation device. Hasani and Freddi (2023) describe Operational Modal Analysis (OMA) as a practical framework for extracting modal information from output measurements under natural or operational loading. This makes OMA relevant to field situations in which a bridge cannot be closed for controlled excitation and in which the objective is to understand the bridge response during normal service.

The literature nevertheless distinguishes monitoring from diagnosis. A frequency change may indicate that the dynamic behavior has changed, but a frequency shift alone generally does not identify a unique damage mechanism or severity. Reference-free damage-detection research emphasizes that reliable diagnosis is more difficult when an intact reference state is unavailable and when environmental or operational effects overlap with damage-related changes (Moravvej & El-Badry, 2024). This distinction supports the present study's limitation to frequency monitoring and baseline characterization rather than structural damage classification.

2.2.2 Natural Frequency, Modal Parameters, and Structural Properties

The dynamic properties of a structure are governed by its mass, stiffness, and boundary conditions. In a simplified single-degree-of-freedom representation, the natural

frequency is related to stiffness and mass through the square-root relationship between those parameters. In multi-degree-of-freedom bridges, the same general principle extends to a set of modal frequencies and corresponding mode shapes. Consequently, changes in structural properties may produce measurable changes in modal parameters. This is the fundamental reason that vibration-based SHM can use modal information as an indirect indicator of structural change (Wang et al., 2022; Azhar et al., 2024).

It is important, however, to distinguish natural frequency from resonant response. Natural frequency is an inherent dynamic property associated with a mode of the structure, whereas resonance describes a condition in which excitation strongly amplifies the response near a modal frequency. In field measurements, the prominent spectral peaks observed under operational excitation are therefore interpreted as candidate modal or dominant frequency components rather than as an automatic measurement of every theoretical natural frequency. This distinction becomes especially important when the excitation spectrum is uneven, when modes are closely spaced, or when the signal-to-noise ratio is low (Chen et al., 2017; Hasani & Freddi, 2023).

2.3. International Perspective

2.3.1 Operational Modal Analysis and Frequency-Domain Identification

Operational Modal Analysis estimates modal characteristics from structural response measurements while the structure is subjected to ambient or operating loads. For bridges, common output-only identification methods include Peak Picking (PP), Frequency Domain Decomposition (FDD), and Stochastic Subspace Identification (SSI). Hasani and

Freddi (2023) report that PP is the simplest frequency-domain approach and identifies candidate modal frequencies from peaks in response spectra. FDD extends frequency-domain analysis by decomposing the spectral density matrix, allowing more complex modal behavior to be separated. SSI operates in the time domain and can provide robust modal identification and mode shapes, although it is more computationally demanding.

Chen et al. (2017) demonstrated the practical value and limitations of these approaches on an eleven-span concrete bridge under weak ambient excitation. Using inexpensive accelerometers, they identified multiple modal frequencies below 10 Hz despite weak excitation and measurement noise. Their comparison showed that Peak Picking could identify a large number of frequencies efficiently, while SSI produced higher-quality mode-shape information at higher computational cost. This finding is directly relevant to an embedded real-time system: a lightweight peak-picking approach can be reasonable for preliminary or continuous monitoring when computational resources are limited, provided that the limitations of the method are explicitly acknowledged.

The literature further indicates that frequency-domain peak identification is most dependable when modal peaks are reasonably separated and the measured response has adequate signal-to-noise ratio. Closely spaced modes, harmonics, local vibration sources, and noise can create ambiguous peaks or shift the apparent maximum. For that reason, the present study should treat the selected spectral peak as a dominant frequency estimate generated by a defined processing procedure, rather than claiming that simple peak picking is universally equivalent to full modal identification (Hasani & Freddi, 2023; Azhar et al., 2024).

2.3.2 Fast Fourier Transform and Vibration Signal Processing

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the Discrete Fourier Transform and is widely used to transform sampled vibration signals from the time domain into a frequency-domain representation. In SHM, this transformation allows the energy distribution of a measured acceleration signal to be examined across frequency. Once the spectrum is formed, candidate dominant or modal frequencies can be selected from spectral peaks, subject to the resolution and quality of the underlying measurement (Hasani & Freddi, 2023).

Field vibration analysis generally requires preprocessing before FFT-based interpretation. Detrending and removal of offset reduce contamination associated with nonstationary mean components, while filtering can be used when the measurement contains strong out-of-band noise. Sampling frequency also constrains the highest frequency that can be represented without aliasing, and the record length affects the spacing of discrete frequency bins. These considerations make the signal-processing chain an integral part of the monitoring method rather than a secondary implementation detail. Chen et al. (2017) and Hasani and Freddi (2023) both show that preprocessing and modal-identification choices affect the reliability of frequency estimates obtained from in-situ bridge responses.

For an embedded system, FFT processing must also be balanced against processor capacity, memory, latency, and the desired update rate. Low-cost edge-computing SHM systems demonstrate that spectral or modal processing can be performed close to the sensor node, reducing the amount of raw data that must be transmitted and enabling near-real-time

monitoring (Hidalgo-Fort et al., 2024). This supports the use of an ESP32-S3-based architecture in which vibration acquisition and frequency extraction occur locally before selected results are sent to a web interface.

2.3.3 Low-Cost MEMS Accelerometers for Bridge Monitoring

The increasing availability of MEMS accelerometers has reduced the cost and physical size of vibration-sensing hardware. Research on low-cost accelerometer systems indicates that such sensors can be used for bridge vibration measurement and eigenfrequency estimation when their noise, sampling behavior, synchronization, and installation conditions are adequately considered. Komarizadehasl et al. (2022a) developed a low-cost triaxial accelerometer system and demonstrated its use for eigenfrequency analysis of a footbridge, while later work by the same research group showed that low-cost wireless accelerometers could support automated field data collection and operational modal analysis (Komarizadehasl et al., 2022b).

More recent field evidence continues to support the feasibility of inexpensive MEMS sensing while emphasizing its limitations. Giermann et al. (2025) tested low-cost MEMS acceleration sensors both under laboratory conditions and on a bridge, where a microcontroller and wireless communication were used to make measurements available online. The study also observed random and systematic influences that caused measurement deviations. These findings reinforce the need for repeatability checks, stable installation, and transparent reporting of sensor limitations when a low-cost platform is used in place of a calibrated commercial monitoring system.

The recent literature also shows that MEMS sensing is moving beyond laboratory prototypes toward practical bridge deployments. Hasani et al. (2024) described a wireless data acquisition system for bridge OMA using MEMS accelerometers and a web-based interface, while Giermann et al. (2025) reported long-term bridge deployment of a low-cost MEMS system. Together, these studies support the feasibility of a compact sensor-microcontroller architecture for field monitoring, but they also highlight that inexpensive sensing should be evaluated in terms of the specific frequency band, noise floor, sampling rate, synchronization, and installation configuration required by the target bridge.

Field studies show that measurement conditions matter when interpreting bridge vibration. Park et al. (2015) reported ambient-vibration field tests using their sensor-node system on large civil infrastructure, including bridge applications. Meng et al. (2019) examined bridge response during vehicle–bridge interaction and described spectral features influenced by the coupled bridge and vehicle dynamics. Camara et al. (2017) used measured pavement profiles to study how surface irregularity affects traffic-induced bridge vibration. These studies support documenting the fixed roadway-deck station, sensor arrangement, traffic context, and road-surface conditions; none determines the best sensor location for Buntun Bridge. Accordingly, this study uses the roadway-deck point marked on Figure 2 as an approximate reference and requires its surveyed position and approved installation to be recorded before formal sessions. The selected FFT peak is treated as a local spectral indicator, not as full modal identification or diagnosis.

2.3.4 Real-Time, Wireless, and IoT-Enabled SHM

Real-time SHM increasingly combines sensing, embedded computation, wireless communication, and web-based visualization. Such architectures allow sensor nodes to preprocess measurements locally and transmit summaries or selected features to a remote interface. Hidalgo-Fort et al. (2024) presented a low-cost, low-power edge-computing SHM architecture for bridges in which processing is performed at the monitoring node and the wireless network is used for communication. The system was deployed on the Eduardo Torroja Bridge and produced modal-frequency results with an error below 1.72% when compared with calibrated commercial systems.

Hasani et al. (2024) similarly integrated MEMS sensing, wireless communication, synchronization, data preprocessing, and a web-based graphical interface into a bridge monitoring architecture. These studies establish that the monitoring problem is not limited to sensor measurement alone; reliable SHM requires a complete data path from acquisition through processing, transmission, storage, and visualization. The present study adopts this system-level perspective, but limits its core analysis to vibration-frequency characterization rather than full-scale automated diagnosis.

A 2026 systematic review by Chaudhari et al. (2026) further identifies low-cost MEMS sensing, IoT connectivity, and data-driven modal analysis as important directions in bridge SHM. At the same time, the review emphasizes that operational and environmental variability remains a major challenge because it can conceal or mimic structural changes. This reinforces the need for the present study to establish its frequency baseline under

explicitly documented operating conditions rather than relying on an undifferentiated pool of field readings.

2.3.5 Environmental and Operational Variability in Frequency Monitoring

One of the most important findings across bridge SHM literature is that modal frequency is not perfectly constant during service. Environmental and operational effects such as temperature, humidity, wind, vehicle loading, traffic intensity, and changing boundary conditions can alter the measured modal characteristics. Sohn (2007) established environmental and operational variability as a central issue in SHM, and Wang et al. (2022) later reviewed methods for predicting and removing these effects from modal-frequency measurements.

Traffic is especially relevant to the present study because it is both the primary operational excitation of the target bridge and a source of variability in the measured response. Zhou and Sun (2019), using long-term monitoring of the Donghai Bridge, demonstrated that traffic and temperature contribute to modal-frequency variation and that such effects can mask changes associated with deterioration. Earlier studies likewise documented changes in measured bridge dynamic characteristics associated with vehicle mass and parked vehicles (Kim et al., 2003; Chang et al., 2014). Thus, differences in traffic condition should not simply be treated as unwanted noise; they are an important operational variable that must be characterized when establishing a field baseline.

For this study, traffic conditions will be represented by three locally defined vehicle count-rate categories: NEAR_ZERO, MEDIUM, and HEAVY. The researchers will select a fixed observation-window length and define numerical cutoffs from pilot traffic counts,

record the choices, and apply them before formal data collection. These categories support consistent grouping of naturally observed sessions; they are not universal civil-engineering traffic classifications. The resulting baseline is specific to the monitored bridge, category definitions, and study period, not an immutable structural constant.

2.4. National/Local Perspective

2.4.1 Philippine Applications of Sensor-Based Bridge SHM

Philippine research demonstrates that sensor-based SHM and remote bridge monitoring are feasible in local infrastructure settings. Gaviña et al. (2017) developed and applied SmartBridge sensor nodes to existing concrete bridges with AASHTO Type IV girders. The system used acceleration measurements, wireless communication, and an online platform for monitoring, with emphasis on appropriate sensor installation and placement. This study is significant to the present research because it shows an earlier Philippine effort to move bridge assessment beyond periodic visual inspection toward instrumented monitoring and remote data access.

Payawal et al. (2017) investigated MEMS-based accelerometer measurements through remote monitoring of the Padre Jacinto Zamora Flyover and compared actual sensor readings with theoretical structural-model outputs. Their work illustrates a practical issue also relevant to the present study: field measurements from low-cost sensors may differ substantially from idealized analytical predictions, making calibration and field characterization necessary rather than assuming that a theoretical value is automatically equivalent to an observed one.

Lau et al. (2018) extended sensor-based bridge monitoring in the Philippine context through a fuzzy-based structural health rating procedure for Zamora Bridge. Three-dimensional acceleration measurements were obtained from sensors installed on structural components and used in an assessment framework. Although the analytical approach differs from the present study, the work demonstrates that local bridge SHM research has already explored acceleration sensing as a supplement to visual inspection and that sensor-derived measurements can be incorporated into condition-assessment workflows.

2.4.2 Philippine Real-Time Monitoring and Institutional Developments

The Philippine Council for Industry, Energy and Emerging Technology Research and Development (PCIEERD) has also supported the Universal Structural Health Evaluation and Recording System (USHER), a system combining accelerographic sensing, a web portal, and a mobile application. The project describes 24/7 monitoring and natural-frequency recording intended to support structural assessment before, during, and after strong earthquakes. It also identifies the limitations of visual inspection as a primary monitoring practice. The existence of this project indicates that real-time, web-connected structural monitoring is already being pursued in the Philippines, although its principal focus and sensing architecture differ from the present bridge-frequency baseline study (PCIEERD, n.d.).

Recent conditions surrounding Buntun Bridge further strengthen the practical relevance of repeated data acquisition. Reporting documented a 15-ton restriction for heavy trucks, while the Philippine Information Agency described fatigue in some steel members and cracks in the concrete deck during the bridge's continuing rehabilitation (Philippine

News Agency, 2025c; Philippine Information Agency, 2025). These reports do not establish a vibration-based diagnosis and should not be interpreted as evidence of a frequency anomaly. Their relevance to this study is contextual: Buntun Bridge is an operating transportation structure for which additional quantitative observations of dynamic behavior may be useful as a supplementary monitoring layer.

2.4.3 Local Research Gap and Relevance to Buntun Bridge

The Philippine studies reviewed above show a progression from acceleration sensing and wireless data transmission toward remote and real-time monitoring. However, the specific combination proposed in the present study remains distinct in emphasis: an embedded, low-cost MEMS-based system is used to obtain field vibration data on Buntun Bridge, transform the measurements into the frequency domain, extract dominant frequency values for repeated sessions, and establish an empirical baseline while explicitly considering traffic conditions. The reviewed local works primarily emphasize acceleration measurement, sensor placement, theoretical calibration, fuzzy-based assessment, or earthquake-oriented monitoring rather than a traffic-conditioned dominant-frequency baseline for an aging steel truss bridge.

This gap should be stated carefully. The literature review does not establish that no other Philippine project has ever performed traffic-conditioned frequency monitoring. Rather, the sources identified for this study do not provide a directly equivalent field methodology combining low-cost MEMS sensing, embedded FFT-based dominant-frequency extraction, repeated traffic-condition sessions, and an empirical baseline for Buntun Bridge. The present study therefore contributes an applied field implementation that

connects established SHM concepts with a resource-conscious monitoring architecture suited to a specific Philippine bridge.

2.5. Synthesis

The reviewed literature establishes four closely connected findings that guide the present study. First, vibration-based SHM is a recognized approach for observing changes in bridge dynamic behavior, with modal frequency among the commonly used features for structural characterization (Das et al., 2016; Azhar et al., 2024). Second, operational modal analysis allows modal information to be estimated from responses under traffic and other ambient loads, making field deployment practical when controlled excitation is not available (Chen et al., 2017; Hasani & Freddi, 2023). Third, low-cost MEMS accelerometers and embedded wireless systems have demonstrated practical potential for bridge monitoring, although noise, synchronization, installation, and calibration remain important limitations (Komarizadehasl et al., 2022a; Komarizadehasl et al., 2022b; Hasani et al., 2024; Giermann et al., 2025). Fourth, traffic and environmental conditions can change measured modal frequencies sufficiently to obscure structural changes, making condition-aware baseline construction essential (Sohn, 2007; Zhou & Sun, 2019; Wang et al., 2022).

The present study uses FFT-based peak selection to produce a dominant-frequency indicator for each onboard channel-axis series at the fixed roadway-deck station. Repeated 10-minute field sessions will be grouped using locally defined NEAR_ZERO, MEDIUM, and HEAVY vehicle-count bands derived from pilot observations. Each session's traffic count and category will be retained with the vibration data. The empirical baseline describes

the observed response at the marked station during the study period; it is not an absolute structural-safety threshold or a traffic-load class.

The literature also supports a clear boundary for interpretation. A statistically significant difference among traffic conditions would demonstrate that operating condition is associated with changes in the measured dominant frequency, but it would not by itself prove structural damage. Likewise, a frequency value outside the baseline would justify attention or further investigation, but it would not identify a specific defect without additional evidence. This conservative interpretation is consistent with the broader SHM literature, which treats environmental and operational effects as sources that must be separated from damage-related changes before making structural-health conclusions (Wang et al., 2022; Chaudhari et al., 2026).

Overall, the reviewed studies justify the development of a localized real-time frequency-monitoring system using MEMS sensors, embedded processing, FFT-based spectral analysis, and web visualization. They also reveal the applied research need addressed by this thesis: documenting a repeatable procedure for recording roadway-deck vibration at a fixed station on an operating Philippine bridge, extracting dominant spectral-frequency indicators, and describing a traffic-aware baseline. This synthesis provides the basis for the research design, data-gathering protocol, repeatability checks, baseline construction, and statistical comparison presented in Chapter III.

CHAPTER III

METHODOLOGY

This chapter presents the methodology for developing and field-evaluating a localized real-time frequency-monitoring system for roadway-deck vibration. The proposed instrument is one PCB carrying an ESP32-S3 and three co-located triaxial IMU ICs at a fixed station. Ten-minute field sessions pair sequential vibration acquisition with a concurrent visual traffic tally; ten 60-second windows per session provide the FFT analysis records. The procedure evaluates acquisition, timing, repeatability, and visualization while remaining within the stated scope of frequency characterization rather than structural diagnosis or load-capacity assessment.

3.1 General Overview

The study employs a developmental and quantitative repeated-session field-testing approach. The developmental component covers the design and integration of one sensing PCB, embedded acquisition and processing, wireless communication, and web visualization. Field testing evaluates whether the system can acquire complete timestamped records and produce repeatable dominant-frequency indicators at a fixed roadway-deck station under naturally occurring traffic-count conditions.

The methodological sequence consists of: (1) securing site permission and safety clearance; (2) bench-checking the single PCB, USB-C power/programming path, and three onboard IMU channels; (3) installing that board at the approved, predeclared roadway-deck station in Figure 2; (4) collecting fixed 10-minute sessions while a separate observer tallies

vehicle passages; (5) dividing each session into ten 60-second analysis windows and checking acquisition timing; (6) processing each channel-axis window with the fixed FFT peak-selection rule; (7) classifying the session from count cutoffs frozen after pilot work; (8) summarizing valid window estimates at the session level and constructing descriptive baselines; and (9) evaluating record completeness, timing, repeatability, and dashboard output. The traffic cutoffs, sampling settings, and site details will be documented before formal collection.

3.2 Study Area

The study will be conducted on Buntun Bridge in Tuguegarao City, Cagayan. Data collection is planned at one predefined roadway-deck station indicated by the red marker in Figure 2, an approximate map reference near the bridge crossing midpoint. The single sensing PCB will remain at this station for every comparison session. Before formal collection, the station's surveyed chainage, lane and transverse offset, board mounting method, and X/Y/Z orientation will be recorded on a site sketch and retained across sessions.

The red marker identifies the proposed roadway-deck station but is not a surveyed coordinate. One PCB carrying all three IMUs will be attached only after the responsible bridge authority approves the exact safe mounting point, lane offset, attachment method, and work procedure. The site sketch and photographs will document the approved board position and axis orientation. If the marked station is not approved as safe or accessible, a replacement roadway-deck station will be selected, mapped, and fixed before formal

collection; the board will not be moved between comparison sessions. Written authorization for access, measurement, and safety procedures is required before deployment. Verbal coordination is not final authorization.

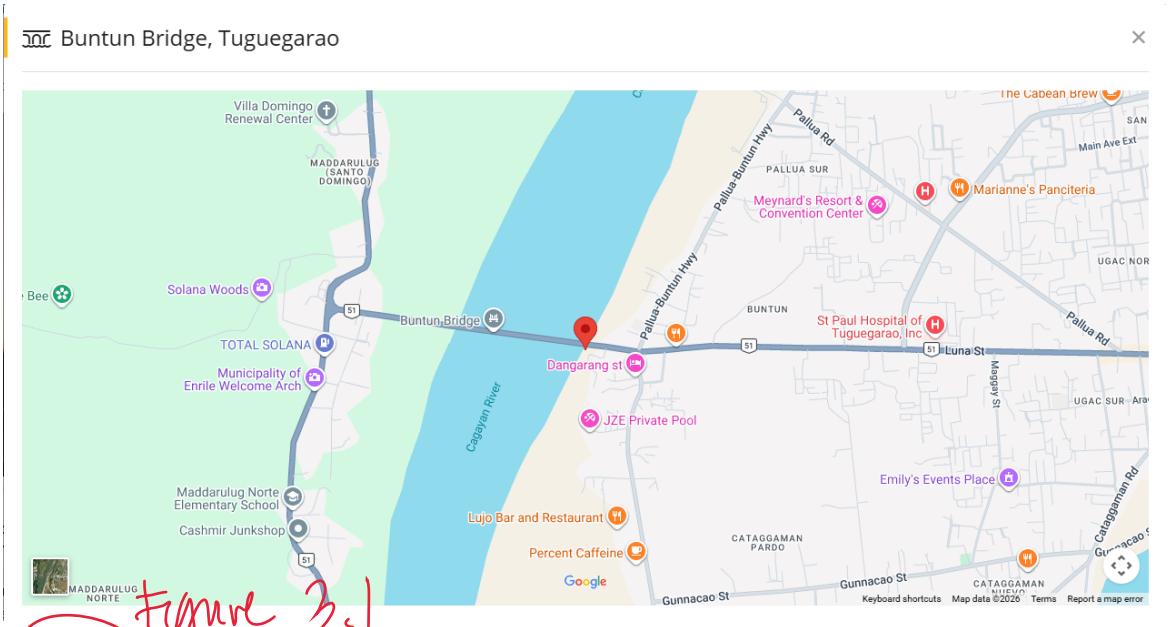


Figure 2. Buntun Bridge study site; the red marker indicates the planned roadway-deck measurement station (approximate map reference).

3.3 Research Design

The study uses a single-site, repeated-session observational field design. Each 10-minute session is one measurement occasion in which the three co-located onboard IMU channels record roadway-deck vibration from the same fixed station, board orientation, scan order, and acquisition settings. Each channel-axis stream is divided into fixed 60-second FFT windows; the median of valid window estimates is the session-level response. Traffic category is assigned from the concurrent vehicle count, not experimentally imposed.

The field design is observational with respect to traffic excitation because the researchers do not control the number, type, speed, or weight of vehicles crossing the bridge. Instead, the study records the bridge response produced by naturally occurring operational loading. This approach is consistent with output-only or operational modal testing, in which vibration responses are collected while a structure remains in service (Hasani & Freddi, 2023).

Because the study does not introduce controlled damage or a known reference excitation, the results are interpreted as an empirical description of frequency behavior for the monitored bridge and study period. A measured frequency outside the established baseline is treated as an observation requiring attention or further investigation, not as proof of a particular damage mechanism.

3.4 System Architecture and Materials

The preliminary monitoring design is one rigid PCB carrying one ESP32-S3 and three co-located MPU-6050 triaxial IMU ICs. The single board is mounted at the fixed roadway-deck station. Its USB-C receptacle accepts the external field supply and provides a USB programming connection to the ESP32-S3; USB is not the planned field dashboard data path. Arduino IDE firmware sequentially polls channel 1 X/Y/Z, channel 2 X/Y/Z, and channel 3 X/Y/Z, tagging readings with channel, axis, sample index, timestamp, and scan-cycle ID. The cycle ID links sequential records and does not indicate simultaneous acquisition. Each channel-axis series is processed in 60-second windows; session summaries and metadata are sent by the separate wireless connection to the web interface.

The hardware labels follow the preliminary schematic; the final assembled board and channel-addressing behavior must be verified on the bench before field use (Espressif Systems, n.d.; InvenSense, 2013).

what do you mean ?

The final manuscript shall report the confirmed board revision, MPU-6050 model, surveyed roadway-deck station, board orientation, lane offset, scan timing, measured per-channel sampling rate, 10-minute session duration, 60-second FFT window, analysis band, external USB-C supply, wireless communication and storage methods, Arduino IDE version, board package, and library versions used in completed trials. Values that require bench verification, site approval, or pilot selection will be finalized and recorded before formal collection rather than inferred from the preliminary schematic.

Component	Purpose and required documentation
Single sensing PCB	One rigid board carrying the ESP32-S3 and three co-located MPU-6050 triaxial IMU ICs; record board revision and channel identifiers.
ESP32-S3	Poll and timestamp the three onboard channels sequentially; process 60-second windows. Record Arduino IDE version, measured per-channel sampling rate, and FFT settings.
Board mounting	Mount the single PCB at the approved fixed roadway-deck station. Record surveyed location, attachment method, and X/Y/Z board orientation.
USB-C and wireless path	Use USB-C for the external field supply and ESP32-S3 programming. Use the separate wireless connection for field summaries. Record the power source and data path.
Web interface	Display frequency results and system status. Record displayed fields and refresh interval.
Field log	Record session ID, date/time, total count and rate for the fixed 10-minute session, pilot-defined traffic group, weather, board station/orientation, and unusual events.

Table 1. *Required documentation of the final monitoring-system configuration*

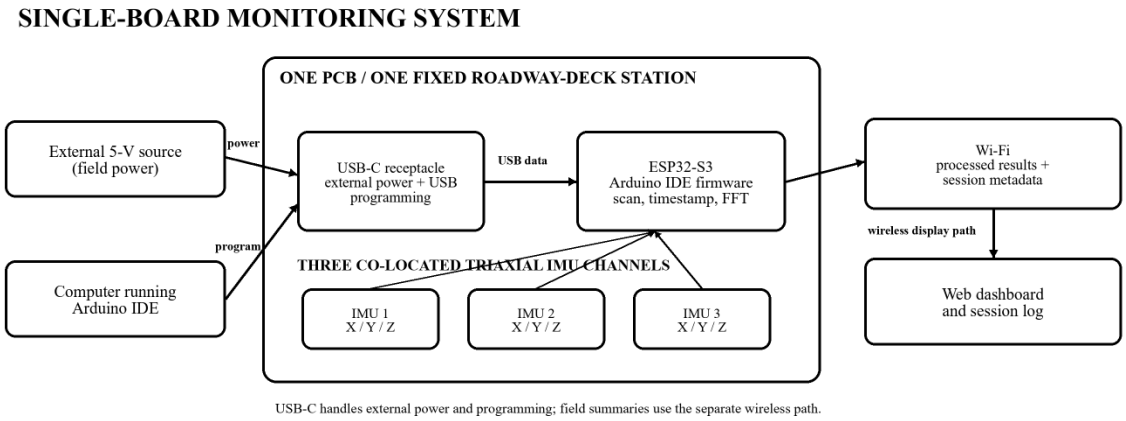


Figure 3. Single-Board Monitoring System Block Diagram

3.5 Field Measurement Units and Sampling Plan

The unit of observation is one valid 10-minute measurement session. A session contains timestamped records from all three co-located IMU channels, polled in the same order at the fixed station and board orientation. The vibration stream is divided into ten consecutive 60-second analysis windows. Each channel-axis window is processed separately; the median of its valid window-level dominant-frequency estimates is retained as the session-level response. One-minute windows are a documented precedent in ambient bridge vibration analysis, but their adequacy for this sensor, frequency band, and field site will be checked during the pilot (Oth & Picozzi, 2012).

The field plan targets five valid 10-minute sessions in each of the three local traffic groups—NEAR_ZERO, MEDIUM, and HEAVY. This is a practical completion target carried forward from the project plan, not a statistical power calculation or a claim of sufficient sample size. A session may be rejected only for predefined technical reasons such as board

movement, severe clipping or saturation, corrupted data, missing timestamps, failed channel acquisition, or timing irregularity beyond the pilot-set tolerance. If a group target cannot be reached within approved field access, report the actual number of valid sessions and keep comparisons descriptive or exploratory.

Session metadata will include date, start and end time, session ID, total vehicle count and rate for the fixed 10-minute window, assigned pilot-derived group, weather and roadway-surface notes, the surveyed station and board orientation, each channel's status, measured sampling intervals, and unusual events. The same board, station, poll order, and processing settings will be retained across sessions.

3.6 Research Instrument and Sensor Installation

The single-board monitoring unit is the primary research instrument. Before deployment, the external USB-C power path, USB programming connection, ESP32-S3, each of the three MPU-6050 channels, sequential poll order, timestamp and scan-cycle continuity, per-channel sampling intervals, raw-data recording, FFT output, and web-interface update will undergo functionality checks. The Arduino IDE firmware and library configuration will be recorded and held constant during field sessions.

The complete PCB, with all three IMU ICs co-located on the board, shall be installed at the approved roadway-deck station shown in Figure 2 using a mechanically stable attachment method authorized for the site. Record the single board's location and X/Y/Z orientation; do not describe the three onboard ICs as sensors distributed across the bridge. Keep the mounting and orientation unchanged across sessions. The installation must

provide stable coupling to the roadway deck while preventing movement and avoiding interference with bridge operations.

A zero-offset or static-bias check shall be conducted before each field run, and the device shall be inspected for looseness or cable movement. Where possible, a short stationary record shall be collected before acquisition to establish whether a large offset, clipping, or abnormal noise is present. These checks support repeatability but do not constitute calibration against a commercial reference instrument.

3.7 Data Gathering Procedure

Formal data collection shall begin only after written permission and site-safety arrangements have been obtained. The researchers shall coordinate with the responsible authority regarding the mapped roadway-deck station, exact lane offset, allowable working locations, traffic restrictions, attachment method, and emergency procedures. No activity shall obstruct traffic lanes or require researchers to enter an unsafe traffic path.

1. Pre-field preparation. Connect the external supply through USB-C, verify the Arduino IDE firmware and recorded library configuration, check all three onboard IMU channels, confirm the poll order and acquisition settings, synchronize the system clock, and prepare the session log and handheld vehicle counter.
2. Site inspection. Verify the approved roadway-deck station against the site sketch, identify safe positions for the board operator and the separate traffic

observer, document roadway-surface and environmental conditions, and confirm the authorized attachment procedure.

3. Board installation. Attach the single PCB securely at the approved roadway-deck station, record its location and orientation, and verify that the three onboard IMU axes match the stated analysis directions.
4. Stationary check. Record a short stationary segment from each sensor and verify that the data are free from obvious saturation, discontinuity, abnormal noise, or missing samples before the session.
5. Concurrent traffic observation. During the same fixed 10-minute session, a separate observer at a safe viewpoint visually counts vehicles crossing the selected reference point with a handheld tally counter. Record the start/end time and total count; assign the session's traffic group only after applying the pilot cutoffs frozen ~~before formal collection~~.
6. Vibration acquisition. Record for exactly 600 seconds while polling channel 1 X/Y/Z, channel 2 X/Y/Z, and channel 3 X/Y/Z sequentially. Timestamp and tag each reading. Divide the acquisition into ten consecutive 60-second analysis windows; do not extend a session to reach a target vehicle count or change the station, board orientation, scan order, or software settings.
7. Session closure. Stop the vehicle tally and vibration acquisition together at 600 seconds. Save the raw timestamped records, channel and axis identifiers, sample indices, scan-cycle IDs, total vehicle count, count rate, traffic group, field-log

→ Show also in table form

metadata, and valid window-level frequency outputs. Inspect the record against the predefined technical acceptance criteria and log any exclusion reason.

8. Repetition. Continue with the same fixed protocol until the practical target of five valid sessions has been reached in each pilot-defined traffic group, subject to approved site access. If a group remains short, retain and report the actual count; do not change the vehicle cutoffs after reviewing frequency results.

3.8 Traffic-Condition Characterization

Traffic condition is represented by the total number of vehicles crossing the fixed reference point during each 10-minute session. FHWA defines traffic volume as vehicles passing a point during a predefined interval and documents 10-minute data increments; this project adopts a fixed 10-minute interval for comparability, not as a universal optimum (Federal Highway Administration [FHWA], 2022). During a pilot, record equal-duration counts across the field periods expected to be sampled. Let m be the number of pilot sessions, and order their vehicle counts from smallest to largest. Define the two nearest-rank empirical tertile cutoffs as shown below. The exact integer cutoffs cannot be supplied until local pilot observations exist.

$$c_1 = C_{([m/3])}, \quad c_2 = C_{([2m/3])}$$

For each fixed 10-minute session, let C be the number of vehicles counted; the observed rate is $r = \frac{C}{T}$ vehicles per minute, with T = 10 minutes. Retain C, the rate, and the assigned local group in the session log. These labels describe traffic volume only; they are not civil-engineering load classes or calibrated measures of structural excitation.

Classify a session as NEAR_ZERO when its count C is at or below the lower cutoff c1, MEDIUM when $c1 < C \leq c2$, and HEAVY when C exceeds the upper cutoff c2. NEAR_ZERO means the lowest pilot-observed local count band and may include nonzero counts. If ties make the cutoffs equal or leave a group empty, repeat or adjust the pilot plan and freeze revised cutoffs before collecting formal sessions. Keep the assigned label, count, and rate in the session record; never reclassify after looking at frequency outcomes.

The traffic labels are descriptive local groupings, not civil-engineering traffic classes, vehicle-weight estimates, or calibrated measures of structural excitation. FHWA describes manual counts as visual observations recorded on a tally sheet or electronic counter; use a handheld counter and a safe observation position. If the pilot shows that simultaneous traffic cannot be counted reliably from one viewpoint, document the limitation and revise the observer arrangement before formal sessions (FHWA, 2013).

The analysis uses all three planned traffic groups. If a group lacks valid sessions or distinct pilot cutoffs, report that limitation and keep results descriptive or exploratory. Do not stop an individual session when it reaches a vehicle-count target; the timer controls its duration, while the pilot-frozen count cutoffs determine the group.

Session field	Required information and use
Session/time	Record session ID and start/end time for the fixed 10-minute observation and vibration-acquisition period.
Traffic group	Assign NEAR_ZERO, MEDIUM, or HEAVY from the two integer cutoffs frozen after pilot work; retain the cutoff version.
Traffic count	Record the visually tallied vehicle count C and rate C/(10 min) while the board records vibration.
Environment	Record weather; add temperature/humidity when available.
Sensor/scan	Record onboard channel, station, axis, sequential scan order, timestamp, sample index, scan-cycle ID, and board orientation.
Frequency output	Record each 60-second channel-axis estimate, its measured sampling rate and N; retain the median valid window estimate as the session value.

Table 2. *Minimum field information recorded for each measurement session*

3.9 Signal Processing and Dominant-Frequency Extraction

Process each accepted 60-second analysis window separately for every onboard channel-axis. Treat its timestamped samples as a uniformly spaced sequence only when the measured timing tolerance is satisfied; remove the window mean, apply the fixed pilot-selected window function, compute the DFT using an FFT, and form the magnitude spectrum. Exclude the zero-frequency bin and select the largest-magnitude bin inside the fixed analysis band. Do not interleave channel records. Apply the same window duration, analysis band, and peak-selection rule to every session.

The Fast Fourier Transform computes the Discrete Fourier Transform efficiently. For each uniformly sampled channel-axis record with measured sampling frequency f_s and N samples, the frequency-bin spacing is given in Equation (1). For a 60-second record with uniform timing, the nominal spacing is 1/60 Hz, approximately 0.0167 Hz; the actual

sample rate, N, timing tolerance, and resulting resolution will be checked during the pilot. Use each channel's measured sampling frequency rather than the overall three-channel scan rate.

$$\Delta f = \frac{f_s}{N} \tag{1}$$

Here, Δf is the frequency-bin spacing in hertz, f_s is the per-channel sampling frequency in hertz, and N is the number of samples in one analysis window. Each FFT input covers 60 seconds; the pilot will verify that timestamp-derived sampling and N yield the required frequency resolution within ESP32-S3 processing and memory limits. The analyzed frequency range will not exceed the Nyquist frequency $\frac{f_s}{2}$.

For the mean-removed and windowed sequence, the complex DFT is:

$$X[k] = \sum_{n=0}^{N-1} w[n](x[n] - \bar{x})e^{-\frac{j2\pi kn}{N}}$$

$$f_k = k\left(\frac{f_s}{N}\right) \tag{2}$$

For real-valued acceleration data, retain the positive-frequency bins up to the highest bin at or below the Nyquist frequency, excluding zero frequency. Restrict selection to the fixed analysis band $0 < f_{\min} \leq f_k \leq f_{\max} \leq \frac{f_s}{2}$. Within that band, select the bin with the greatest magnitude. The selected bin index is k^*. The dominant-frequency estimate is then $f_d = \frac{k^* f_s}{N}$; no sub-bin interpolation is applied. The same per-sensor-axis band and selection rule will be used for every session.

Peak picking is computationally lightweight and is suitable for a constrained embedded implementation when the dominant peaks are sufficiently separated. However, because peak picking can be ambiguous for closely spaced modes or weak/noisy responses, the result is treated primarily as a repeatable dominant spectral-frequency indicator. It is not automatically assumed to represent every theoretical natural frequency or a unique structural mode (Hasani & Freddi, 2023).

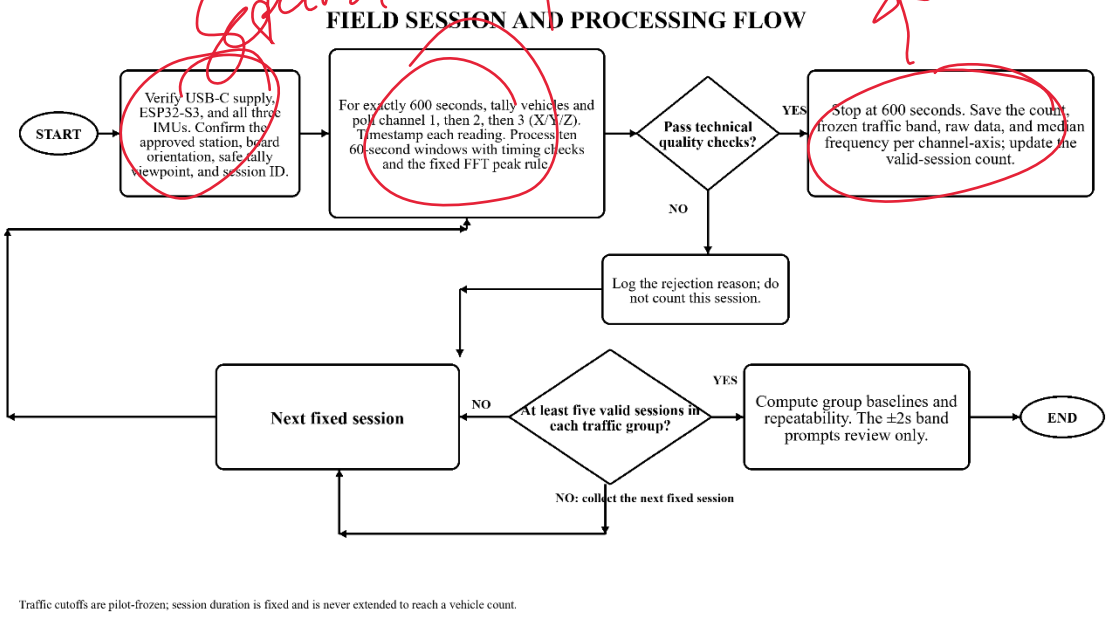


Figure 4. Field Data Collection and Processing Flowchart

3.10 Baseline Frequency Construction

For each traffic group, onboard channel, and axis, construct the baseline from valid session-level estimates, where each session estimate is the median of its valid 60-second window estimates. The baseline is internal to this device, one roadway-deck station, and study period; it is not an accepted safety limit or an externally validated reference.

For one traffic-category, sensor, and axis stratum, let f_i denote the dominant-frequency estimate from valid session i, with $n \geq 2$ sessions. The sample mean and sample standard deviation are calculated as follows (NIST/SEMATECH, n.d.-a):

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i}{n}$$

$$s = \sqrt{\frac{\sum_{i=1}^{n(f_i - \bar{x})^2}}{n - 1}}$$

$$Baseline\ interval = \bar{x} \pm s \tag{3}$$

Here, $\bar{x}$ is the mean dominant-frequency estimate and s is the sample standard deviation within that traffic-category, sensor, and axis stratum. The descriptive baseline interval is shown in Equation (3). The coefficient of variation is also reported to describe relative dispersion:

$$CV = \left(\frac{s}{\bar{x}}\right) \times 100\% \tag{4}$$

The interval in Equation (3) describes the observed one-standard-deviation spread across session-level estimates. A separate configurable review band is $\bar{x} \pm k_{(review)s}$, with $k_{review} = 2$ as the default. Values outside this band are flagged for review only. The default two-standard-deviation review band is a project heuristic, not a 95% confidence or prediction interval, structural-safety limit, or damage threshold; formal prediction limits depend on sample size and a specified statistical model (NIST/SEMATECH, n.d.-c). The final thesis shall report session-level n, traffic cutoffs and counts, channel/axis identity, and the chosen review multiplier for every baseline.

3.11 Statistical Treatment

Descriptive statistics will summarize the session-level dominant-frequency estimates separately for each traffic category, sensor, and axis. Report valid session count n, mean, sample standard deviation, minimum, maximum, and coefficient of variation. Sensor and axis readings collected within one session are linked measurements and will not be counted as independent session replicates.

The primary analysis is descriptive. For each traffic group, onboard channel, and axis, report session-level n, mean, median, sample standard deviation, minimum, maximum, and coefficient of variation. The five-session-per-group target is a practical completion target, not evidence of adequate statistical power or sample size. Before any confirmatory category comparison, calculate and document the required sample size using a prespecified effect of interest, significance level, desired power, and a defensible variance estimate. NIST/SEMATECH notes that sample-size requirements depend on significance level, power, and population variation (NIST/SEMATECH, n.d.-d). If the available sample is not supported by that calculation, treat category differences as exploratory and do not claim confirmatory significance. For a justified inferential comparison, set the significance level at $\alpha = 0.05$; any post-hoc comparison will be selected and reported with the actual sample sizes and assumptions (NIST/SEMATECH, n.d.-b).

If the prespecified sample-size analysis supports an inferential comparison, use one-way ANOVA separately for each channel-axis response only when observations are independent and its assumptions are reasonably supported. Examine residual behavior and equality of variances using suitable diagnostics; consider diagnostic plots, group sizes,

outliers, and measurement context rather than interpreting a normality test alone. If ANOVA assumptions are not reasonably supported, use Kruskal–Wallis only when category samples are independent. Report actual sample sizes and the basis for the selected analysis.

The three traffic categories are analyzed separately for each sensor-axis; results from different sensors or axes are not pooled as independent observations. If session independence or sample size does not support an inferential comparison, report descriptive results and treat between-category differences as exploratory.

3.12 Reliability and Repeatability Assessment

System performance will be assessed using measures available from the prototype and field logs: session completion and validity; the percentage of required samples/windows present for each channel-axis; measured per-channel sampling interval and timing variability; agreement between the embedded FFT result and an independent offline FFT applied to the same raw window; repeatability of session-level frequency estimates; and successful dashboard/session-log updates. These checks evaluate acquisition and processing performance. Without a calibrated reference accelerometer, they do not establish absolute acceleration or frequency accuracy.

For DSP verification, feed a generated test signal with a known frequency through the same FFT peak-selection settings and verify that the returned bin follows the expected frequency-bin spacing. For field-data comparison, independently recompute the FFT from the exact recorded sample window and compare the selected bin with the firmware output. Under similar mounting and traffic-count conditions, compare repeated session-level

medians using the mean, sample standard deviation, and coefficient of variation. Log board revisions, remounting, firmware, sampling, and processing changes; do not present repeatability as calibration accuracy.

3.13 Data Quality Control and Acceptance Criteria

Admit a session only when the board remained fixed at the approved station, all three onboard IMU channels have complete timestamped records for the required 10-minute session, the fixed scan order and timing tolerance are met, the ten 60-second windows and metadata are present, and the signals are suitable for the predefined FFT procedure. Exclude sessions with saturation, severe clipping, unrecoverable missing or irregular samples, corrupted files, or documented hardware failure, and record the reason in a data-quality log.

The researchers shall avoid selective removal of observations merely because their frequencies are unusually high or low. Outliers will first be investigated for technical causes. If no technical cause is identified, the observation shall remain in the dataset and be addressed through transparent statistical analysis. This procedure reduces the risk of manipulating the baseline to obtain a desired result.

3.14 Ethical, Safety, and Site Considerations

The study involves field work on an active bridge and therefore requires safety procedures beyond those applicable to laboratory testing. Written authorization from the responsible authority shall be obtained before formal deployment. Researchers shall remain in designated safe areas, use appropriate personal protective equipment, secure cables and

hardware against movement, and avoid any activity that could distract drivers, obstruct traffic, or compromise bridge operations.

The study will not intentionally impose abnormal or unsafe loads on the bridge. Vehicle activity is observed as it occurs during normal operation. No personally identifying information from drivers or pedestrians is required for the vibration-frequency analysis. If photographs or field videos are collected for documentation, they shall be limited to what is necessary for the methodology and shall avoid unnecessary identification of individuals.

3.15 Summary of the Methodological Framework

The methodology links vibration response at one fixed roadway-deck station to system output through repeated 10-minute sessions. One PCB carries three co-located triaxial IMUs. Firmware polls channel 1 X/Y/Z, channel 2 X/Y/Z, and channel 3 X/Y/Z sequentially; a separate observer tallies vehicle passages during the same 600 seconds. Each session contains ten 60-second analysis windows. Each valid window is processed separately, and the median valid window estimate gives the session-level result for each channel-axis. Pilot-frozen vehicle-count cutoffs assign the local traffic label. Baselines use independent session-level estimates; x-bar plus/minus s describes spread and a separate default two-s band prompts review only. The system evaluation covers record completeness, timing, DSP agreement, repeatability, and dashboard output; it does not diagnose damage or assess safety.

This methodology is intentionally limited to frequency characterization of roadway-deck vibration at the documented station and system evaluation. It does not establish a structural damage threshold, diagnose a specific defect, determine load capacity, or replace

formal bridge inspection. These boundaries preserve consistency between the methods, the stated objectives, and the scope of the study.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter is reserved for system acquisition and processing performance and per-channel, per-axis frequency results by pilot-defined traffic group. This proposal-stage draft contains no measured field findings.

4.1 Introduction

No measured field results are available at this proposal stage. After approved collection, this section will summarize valid-session counts and exclusions and introduce the system-performance and roadway-deck frequency results.

4.2 System Acquisition and Processing Performance

System-evaluation results will include session completion and validity, channel-axis sample completeness, measured scan timing, embedded-versus-offline FFT agreement, repeatability, and dashboard/session-log updates, based on completed bench and field records.

4.3 Roadway-Deck Frequency by Traffic-Count Group

Frequency results will be summarized by the pilot-frozen NEAR_ZERO, MEDIUM, and HEAVY groups, onboard channel, and axis, with observed vehicle counts and sample sizes. Group differences will be treated as exploratory unless a prespecified sample-size calculation supports inference. These results do not constitute a damage diagnosis or safety determination.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

This chapter will summarize completed procedures and findings and state conclusions and recommendations supported by collected data. It remains to be completed after field collection and analysis.

5.1 Summary

The summary will be completed after field collection and will include the finalized study purpose, approved station, confirmed hardware and software configuration, valid-session counts, and principal descriptive results from the actual study record.

5.2 Conclusion

Conclusions will be drawn from completed results and will address each research objective. They will not claim structural damage diagnosis or bridge-safety assessment.

5.3 Recommendations

Recommendations will follow from the completed evaluation results and may identify development changes or further validation supported by the analysis.

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